Antiderivatives and Indefinite Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find each indefinite integral:

a $\int (3x^2 - 4x + 5)\,dx = x^3 - 2x^2 + 5x + C$

b $\int \frac{1}{x^2}\,dx = -\frac{1}{x} + C$

c $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C$

d $\int \sec^2 x\,dx = \tan x + C$

(a) $= x^3 - 2x^2 + 5x + C$; (b) $= -\frac{1}{x} + C$; (c) $= \frac{2}{3}x^{3/2} + C$; (d) $= \tan x + C$

2 [Foundational] • Target: Basic antiderivatives

Find $\int (4\sin x + e^x)\,dx$.

$= -4\cos x + e^x + C$.

3 [Foundational] • Target: Using initial conditions

Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.

$f(x) = 2x^3 - 2x + C$. Using $f(1) = 5$: $2 - 2 + C = 5$, so $C = 5$: $f(x) = 2x^3 - 2x + 5$.

4 [Foundational] • Target: Using initial conditions

A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

$s(t) = \int v(t)\,dt = t^3 - 3t^2 + C$. Using $s(0) = 4$: $C = 4$: $s(t) = t^3 - 3t^2 + 4$.

5 [Foundational] • Target: Using initial conditions

An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s^2.

a Find the velocity function $v(t)$. $v(t) = -9.8t$ m/s (using $v(0) = 0$).

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$. $s(t) = -4.9t^2 + 45$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places. $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

(a) $v(t) = -9.8t$ m/s (using $v(0) = 0$).; (b) $s(t) = -4.9t^2 + 45$.; (c) $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

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6 [Foundational] • Target: More antiderivatives

Find the general antiderivative of $f(x) = (x+1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

Treating $(x+1)$ as a single unit (power rule for a linear inner function): $\int (x+1)^3\,dx = \frac{(x+1)^4}{4} + C$.

7 [Foundational] • Target: Motion applications

A rocket's acceleration is $a(t) = 6t$ m/s^2, with $v(0) = 0$ and $s(0) = 0$.

a Find $v(t)$. $v(t) = 3t^2$.

b Find $s(t)$. $s(t) = t^3$.

c Find the distance traveled in the first 4 seconds. $s(4) = 64$ m.

(a) $v(t) = 3t^2$.; (b) $s(t) = t^3$.; (c) $s(4) = 64$ m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Antiderivatives of composite expressions

Find $\int (2x-1)^5\,dx$, treating $(2x-1)$ as a linear inner function.

$\int (2x-1)^5\,dx = \frac{(2x-1)^6}{6 \cdot 2} + C = \frac{(2x-1)^6}{12} + C$ (dividing by the derivative of the inner function, 2).

9 [Intermediate] • Target: Antiderivatives of composite expressions

Find $\int \sin(3x)\,dx$ and $\int e^{5x}\,dx$.

$\int \sin(3x)\,dx = -\frac{1}{3}\cos(3x) + C$. $\int e^{5x}\,dx = \frac{1}{5}e^{5x} + C$.

10 [Intermediate] • Target: Verifying antiderivatives by differentiation

Verify that $F(x) = x\ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.

$F'(x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x = f(x)$ ✓.

11 [Intermediate] • Target: A projectile motion application

A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8$ m/s^2. Find its velocity function $v(t)$ and position function $s(t)$.

$v(t) = \int a(t)\,dt = -9.8t + C_1$. Using $v(0) = 15$: $v(t) = -9.8t + 15$.
$s(t) = \int v(t)\,dt = -4.9t^2 + 15t + C_2$. Using $s(0) = 2$: $s(t) = -4.9t^2 + 15t + 2$.

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12 [Intermediate] • Target: Visualizing an antiderivative family

The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F.

a Write the general antiderivative of $f(x) = 2x$. $F(x) = x^2 + C$.

b If a specific antiderivative passes through $(1, 4)$, find its constant C. $1^2 + C = 4$, so $C = 3$: $F(x) = x^2 + 3$.

(a) $F(x) = x^2 + C$.; (b) $1^2 + C = 4$, so $C = 3$: $F(x) = x^2 + 3$.

13 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int \frac{x^3 + 2x}{x}\,dx$ by simplifying the integrand before integrating.

$\frac{x^3 + 2x}{x} = x^2 + 2$, so $\int (x^2 + 2)\,dx = \frac{x^3}{3} + 2x + C$.

14 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int (x + 1)(x - 3)\,dx$ by expanding the product first.

$(x + 1)(x - 3) = x^2 - 2x - 3$, so $\int (x^2 - 2x - 3)\,dx = \frac{x^3}{3} - x^2 - 3x + C$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 12.02 Power Series and Taylor Series

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

Final Answer: $2e^{-1} + 1$. The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r - 1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 te^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$.
$y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 te^t = (2 + c_2)e^t + (2 + c_2)te^t$..

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16 [Challenging] • Target: Using initial conditions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

Final Answer: π. The region R is defined by $y = \frac{1}{x^2+1}$, $y = 0$, $x = 0$, and $x = 1$. We are rotating this region about the y-axis. Using the method of cylindrical shells, the volume V is given by the integral: $V = \int_a^b 2\pi x h(x) dx$ where x is the radius of the cylindrical shell, and $h(x)$ is the height of the shell. In this case, the radius of a cylindrical shell at a distance x from the y-axis is simply x. The height of the shell is the value of the function $y = \frac{1}{x^2+1}$..

17 [Challenging] • Target: More antiderivatives • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:.

Final Answer: $1 + x + \frac{x^2}{2}$. Substituting $u = x - \frac{x^3}{6} + \ldots$ into $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \ldots$ gives $1 + \left(x - \frac{x^3}{6}\right) + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4) = 1 + x + \frac{x^2}{2}$.

18 [Challenging] • Target: Motion applications • Cross-Lesson: 15.01 Double Integrals

Volume of cone: radius r, height h, using disc method:.

Final Answer: $\frac{1}{3}\pi r^2 h$. Applying Applications of Integration principles, $V = \frac{1}{3}\pi r^2 h$. Therefore, the verified correct option is $\frac{1}{3}\pi r^2 h$.

19 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Volume of region $y \geq x^2$, $y \leq 4$ rotated about $y = 4$ using washers:.

Final Answer: $512\pi/15$. Applying Applications of Integration principles, $V = \pi\int_{-2}^{2}((4 - x^2)^2)dx$. Therefore, the verified correct option is $512\pi/15$.

20 [Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 2.01 Evaluating Limits

Error bound of Simpson's rule on $[a, b]$ with n intervals:.

Final Answer: $O(h^4)$. Applying Applications of Integration principles, Simpson: error $= O(h^4)$. Therefore, the verified correct option is $O(h^4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Power Rule for Integration

Evaluate $\int (6x^3 - 4x + 5)\,dx$.

a. $\frac{3}{2}x^4 - 2x^2 + 5x + C$ ✓

b. $2x^4 - 2x^2 + 5x + C$

c. $18x^2 - 4 + C$

d. $\frac{3}{2}x^4 - 4x^2 + 5x + C$

$\int 6x^3\,dx = 6(x^4/4) = \frac{3}{2}x^4$. $\int -4x\,dx = -2x^2$. $\int 5\,dx = 5x$. Result: $\frac{3}{2}x^4 - 2x^2 + 5x + C$.

22 [Foundational] • Target: The Power Rule for Integration

Find $\int \left(x^{2/3} + \frac{1}{x^3}\right) dx$.

a. $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$ ✓

b. $\frac{5}{3}x^{5/3} - \frac{1}{4x^4} + C$

c. $\frac{3}{5}x^{5/3} + \frac{1}{2x^2} + C$

d. $\frac{2}{5}x^{5/3} - \frac{2}{x^2} + C$

$\int x^{2/3}\,dx = \frac{x^{5/3}}{5/3} = \frac{3}{5}x^{5/3}$. $\int x^{-3}\,dx = \frac{x^{-2}}{-2} = -\frac{1}{2x^2}$. Result: $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$.

23 [Foundational] • Target: The Power Rule for Integration

Evaluate $\int \frac{3x^4 - 2x}{x^2}\,dx$.

a. $\frac{3}{3}x^3 - 2\ln|x| + C = x^3 - 2\ln|x| + C$ ✓

b. $x^3 - \frac{2}{x} + C$

c. $3x^3 - 2x + C$

d. $x^3 - 2x^{-1} + C$

Simplify integrand: $\frac{3x^4}{x^2} - \frac{2x}{x^2} = 3x^2 - \frac{2}{x}$. Integrating: $x^3 - 2\ln|x| + C$.

24 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Evaluate $\int (3\sin x + 4\cos x)\,dx$.

a. $-3\cos x + 4\sin x + C$ ✓

b. $3\cos x - 4\sin x + C$

c. $3\cos x + 4\sin x + C$

d. $-3\cos x - 4\sin x + C$

$\int \sin x\,dx = -\cos x$ and $\int \cos x\,dx = \sin x$. Thus $\int (3\sin x + 4\cos x)\,dx = -3\cos x + 4\sin x + C$.

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25 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Evaluate $\int (5e^x + \sec^2 x)\,dx$.

a. $5e^x + \tan x + C$ ✓

b. $5e^x - \tan x + C$

c. $5e^x + \sec x \tan x + C$

d. $e^x + \tan x + C$

$\int e^x\,dx = e^x$ and $\int \sec^2 x\,dx = \tan x$. Result is $5e^x + \tan x + C$.

26 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Find $\int (2^x + \csc x \cot x)\,dx$.

a. $\frac{2^x}{\ln 2} - \csc x + C$ ✓

b. $2^x \ln 2 - \csc x + C$

c. $\frac{2^x}{\ln 2} + \csc x + C$

d. $2^x - \cot x + C$

$\int a^x\,dx = \frac{a^x}{\ln a}$ and $\int \csc x \cot x\,dx = -\csc x$. Result is $\frac{2^x}{\ln 2} - \csc x + C$.

27 [Foundational] • Target: Initial Value Problems

Find $f(x)$ given $f'(x) = 4x - 3$ and $f(1) = 6$.

a. $f(x) = 2x^2 - 3x + 7$ ✓

b. $f(x) = 2x^2 - 3x + 5$

c. $f(x) = 4x^2 - 3x + 5$

d. $f(x) = 2x^2 - 3x$

$f(x) = \int (4x - 3)\,dx = 2x^2 - 3x + C$. $f(1) = 2(1)^2 - 3(1) + C = 6 \Rightarrow -1 + C = 6 \Rightarrow C = 7$.

28 [Foundational] • Target: Initial Value Problems

A curve satisfies $\frac{dy}{dx} = 3x^2 + 2x$ and passes through $(2, 10)$. Find its equation.

a. $y = x^3 + x^2 - 2$ ✓

b. $y = x^3 + x^2 + 2$

c. $y = 3x^3 + 2x^2 - 22$

d. $y = x^3 + x^2$

$y = x^3 + x^2 + C$. $y(2) = 8 + 4 + C = 10 \Rightarrow 12 + C = 10 \Rightarrow C = -2$. Equation: $y = x^3 + x^2 - 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Initial Value Problems

Given $f''(x) = 6x$, $f'(0) = 4$, and $f(0) = 1$, find $f(x)$.

a. $f(x) = x^3 + 4x + 1$ ✓

b. $f(x) = 3x^3 + 4x + 1$

c. $f(x) = x^3 + 2x + 1$

d. $f(x) = x^3 + 4$

$f'(x) = 3x^2 + C_1$. $f'(0) = 4 \Rightarrow C_1 = 4$. $f(x) = x^3 + 4x + C_2$. $f(0) = 1 \Rightarrow C_2 = 1$.

30 [Intermediate] • Target: Rectilinear Motion and Acceleration

A particle moves with acceleration $a(t) = 6t - 4$. If its initial velocity $v(0) = 5$, find $v(t)$.

a. $v(t) = 3t^2 - 4t + 5$ ✓

b. $v(t) = 6t^2 - 4t + 5$

c. $v(t) = 3t^2 - 4t$

d. $v(t) = 3t^2 + 4t + 5$

$v(t) = \int (6t - 4)\,dx = 3t^2 - 4t + C$. $v(0) = 5 \Rightarrow C = 5$. Thus $v(t) = 3t^2 - 4t + 5$.

31 [Intermediate] • Target: Rectilinear Motion and Acceleration

If a ball is thrown upward with $v(0) = 48$ ft/s and $a(t) = -32$ ft/s^2 from ground level $s(0) = 0$, find its maximum height.

a. 36 ft ✓

b. 48 ft

c. 64 ft

d. 32 ft

$v(t) = -32t + 48 = 0 \Rightarrow t = 1.5$ s. $s(t) = -16t^2 + 48t$. At $t = 1.5$:
$s(1.5) = -16(2.25) + 48(1.5) = -36 + 72 = 36$ ft.

32 [Intermediate] • Target: Rectilinear Motion and Acceleration

A car decelerates at a constant $a = -4$ m/s^2 from $v_0 = 24$ m/s. How far does it travel before coming to a stop?

a. 72 m ✓

b. 48 m

c. 96 m

d. 36 m

$v(t) = 24 - 4t = 0 \Rightarrow t = 6$ s. Distance $s(t) = 24t - 2t^2$. At $t = 6$:
$s(6) = 24(6) - 2(36) = 144 - 72 = 72$ m.

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33 [Intermediate] • Target: Initial Value Problems

Area of triangle with vertices at origin, $(3,0)$, $(0,4)$ using integration:

a. 3

b. 12

c. 6 ✓

d. π

Applying Applications of Integration principles, $A = \frac{1}{2}(3)(4) = 6$. Therefore, the verified correct option is 6.

34 [Intermediate] • Target: Rectilinear Motion and Acceleration

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}\,]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π ✓

d. $\sqrt{\pi}$

$V = 2\pi\int_0^{\sqrt{\pi/2}} x\cos(x^2)dx = \pi[\sin(x^2)]_0^{\sqrt{\pi/2}} = \pi\sin(\pi/2) = \pi.$

35 [Intermediate] • Target: Basic antiderivatives

Approximate $\int_0^1 \frac{\sin x}{x}dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$ ✓

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

$\sin x/x = \sum(-1)^n x^{2n}/(2n+1)!\,;\, \int_0^1 = \sum(-1)^n/(2n+1)(2n+1)! = 1 - 1/18 + 1/600 - \cdots.$

36 [Intermediate] • Target: Using initial conditions

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$ ✓

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

$S = 2\pi\int_{-R}^{R}\sqrt{R^2 - x^2}\,\sqrt{1 + x^2/(R^2 - x^2)}\,dx = 4\pi R^2.$

37 [Intermediate] • Target: More antiderivatives

Power series integration: $\int_0^x \frac{1}{1+t^2}dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} =$

a. $\arctan x$ ✓

b. $\ln(1+x)$

c. $\sin x$

d. $x/(1+x^2)$

Term-by-term integration of geometric series $1/(1+t^2)$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Motion applications • Cross-Lesson: 15.01 Double Integrals

Numerical integration: Simpson's rule uses polynomials of degree:

a. 1

b. 3

c. **2** ✓

d. 4

Parabolas (degree 2) for each pair of subintervals.

39 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Volume of a sphere radius R using disc method:

a. $\frac{4}{3}\pi R^3$ ✓

b. $\frac{4}{3}\pi R^2$

c. $4\pi R^2$

d. $2\pi R^3$

$V = \pi\int_{-R}^{R}(R^2 - x^2)dx = \frac{4}{3}\pi R^3$.

40 [Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 2.01 Evaluating Limits

Volume integral: using spherical coordinates, $dV =$

a. $dr\,d\theta\,d\phi$

b. $r^2\sin\phi\,dr\,d\theta\,d\phi$ ✓

c. $r\sin\theta\,dr\,d\theta\,d\phi$

d. $r^2\,dr\,d\theta\,d\phi$

Spherical: $dV = r^2\sin\phi\,dr\,d\theta\,d\phi$.

41 [Challenging] • Target: Antiderivatives of reciprocal and rational-power functions • Cross-Lesson: 2.02 Continuity

LIATE rule for IBP: first choice for u among x^2, e^x, $\ln x$:

a. e^x

b. x^2

c. $\ln x$ ✓

d. any of them

Logarithm comes before Algebraic: $u = \ln x$.

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42 [Challenging] • Target: A projectile motion application • Cross-Lesson: 3.01 The Derivative from First Principles

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m+n$. It typically converges more:

a. slowly than Taylor

b. only at 0

c. **rapidly than Taylor for poles ✓**

d. never

Padé: better than Taylor near poles.

43 [Challenging] • Target: Visualizing an antiderivative family • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2-4)^2 dx$, find the value of $b - a$.

a. 2

b. **4 ✓**

c. 8

d. 16

The region is bounded by the curve $y = x^2 - 4$ and the x-axis. To find the limits of integration, we need to find the x-intercepts of the curve, where $y = 0$. Setting $x^2 - 4 = 0$, we get $x^2 = 4$, which gives $x = \pm 2$. Therefore, the bounds of integration are $a = -2$ and $b = 2$. The volume of the solid of revolution is given by $V = \pi\int_a^b [f(x)]^2 dx$. In this case, $f(x) = x^2 - 4$, and the limits are $a = -2$ and $b = 2$. The question asks for the value of $b - a$. $b - a = 2 - (-2) = 2 + 2 = 4$.

44 [Challenging] • Target: Antiderivatives requiring algebraic rewriting first • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Water drains from a cone-shaped tank (radius 6, height 12) at $\frac{dV}{dt} = 1$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{6}{12}h$, find $\frac{dh}{dt}$ when $h = 5$.

a. $-\frac{4}{25\pi}$

b. $\frac{8}{25\pi}$

c. $\frac{2}{25\pi}$

d. $\frac{4}{25\pi}$ **✓**

$V = \frac{\pi}{3}(\frac{6}{12})^2 h^3 \Rightarrow \frac{dV}{dt} = \pi(\frac{6}{12})^2 h^2 \frac{dh}{dt}$. Solving at $h = 5$: $\frac{dh}{dt} = \frac{4}{25\pi}$ m/min.

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45 [Challenging] • Target: The Power Rule for Integration • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Water drains from a cone-shaped tank (radius 4, height 10) at $\frac{dV}{dt} = 2$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{4}{10}h$, find $\frac{dh}{dt}$ when $h = 2$.

a. $\frac{25}{4\pi}$

b. $-\frac{25}{8\pi}$

c. $\frac{25}{16\pi}$

d. $\frac{25}{8\pi}$ ✓

$V = \frac{\pi}{3}(\frac{4}{10})^2 h^3 \Rightarrow \frac{dV}{dt} = \pi(\frac{4}{10})^2 h^2 \frac{dh}{dt}$. Solving at $h = 2$: $\frac{dh}{dt} = \frac{25}{8\pi}$ m/min.

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